



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Smart ATM Transaction Validation System Using State Machines

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of

CSA 1304 Theory of computation
to the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Smart ATM Transaction Validation System Using State Machines**” has been carried out **Pusalapati Chandu (192372119)** under the supervision of Thiyagarajan A and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, the undersigned students of the Electronics and Communication Engineering department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Smart ATM Transaction Validation System Using State Machines**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

All sources, references, software resources, datasets, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been followed throughout the planning, development, testing, and documentation of the project.

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INDIVIDUAL CONTRIBUTION STATEMENT

The following table records the major responsibilities and contributions of each team member. The exact allocation may be updated by the team if the final implementation differs from the module allocation used in the project presentation.

For the project title “**Smart ATM Transaction Validation System Using State Machines**”, you can use this contribution table:

Student / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution
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Pusalapati Chandu 192372119	Transaction Processing	Withdrawal, deposit and balance enquiry states	Transaction validation and functional testing	Ch. 2, Ch. 4	25%
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ABSTRACT

The Smart ATM Transaction Validation System Using State Machines is a secure and efficient system designed to manage and validate ATM transactions using the concept of Finite State Machines (FSM). The main objective of the system is to provide a structured and reliable transaction process while preventing unauthorized access and invalid operations.

The system represents each stage of an ATM transaction as a separate state. The major states include Card Detection, PIN Entry, PIN Verification, Account Validation, Transaction Selection, Transaction Processing, Transaction Recording, Transaction Completed, and Authentication Failed. The system moves from one state to another based on user inputs and predefined conditions.

When a user inserts a card, the system verifies the card details and requests the PIN. The entered PIN is checked against the account information. If the authentication is successful, the user is allowed to select operations such as cash withdrawal, deposit, balance enquiry, and transaction history. If incorrect credentials are entered, the system provides retry options and can block the session after multiple failed attempts.

The system also handles conditions such as invalid cards, incorrect PINs, insufficient balance, transaction failure, blocked accounts, session timeout, and unauthorized access. Every successful or failed transaction is recorded to maintain a proper transaction history and improve security and accountability.

Using state machines makes the ATM transaction process easier to design, understand, test, and maintain. Each state has clearly defined inputs, outputs, and transitions, reducing unexpected system behavior. Overall, the proposed system improves security, reliability, accuracy, transaction control, and user experience in ATM operations. It demonstrates how state-machine concepts can be effectively applied to develop a secure real-world financial transaction system.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
ATM	Automated Teller Machine	4
PIN	Personal Identification Number	
FSM	Finite State Machine	4
DFA	Deterministic Finite Automaton	1
NFA	Non-Deterministic Finite Automaton	2
UI	User Interface	—
DB	Database	—
ID	Identification / Account Identification	—
T	Transaction	—
A	Account	—
B	Account Balance	Currency
W	Withdrawal Amount	Currency
D	Deposit Amount	Currency
S	System State	—
I	Input	—
O	Output	—
N	Number of PIN Attempts	Count
V	Transaction Validation	—
TID	Transaction ID	—
OTP	One-Time Password	—
Auth.	Authentication	—
Sec.	Security	—
API	Application Programming Interface	—
SQL	Structured Query Language	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Automated Teller Machines (ATMs) are widely used for performing banking transactions such as cash withdrawal, cash deposit, balance enquiry, and account-related services. Since ATM systems handle financial information and transactions, they must provide secure, reliable, and controlled access to authorized users.

A typical ATM transaction involves several sequential activities, including card insertion, card validation, PIN entry, PIN verification, account authentication, transaction selection, transaction processing, and transaction completion. Each activity depends on the successful completion of the previous activity. An error or unauthorized action at any stage should prevent the transaction from continuing.

The **Smart ATM Transaction Validation System Using State Machines** applies the concept of a **Finite State Machine (FSM)** to model and control the different stages of an ATM transaction. Each stage of the transaction is represented as a state, and the system moves from one state to another based on specific inputs and conditions.

The major states include **Card Detected, PIN Entry, PIN Verification, Account Verified, Transaction Selection, Transaction Processing, Transaction Recorded, Transaction Completed, and Authentication Failed**. These states provide a structured mechanism for controlling ATM operations.

For example, when a valid card is inserted, the system moves to the PIN Entry state. If the entered PIN is correct, the system moves to Account Verified and then allows transaction selection. If an incorrect PIN is entered, the system provides an opportunity to retry. After multiple unsuccessful attempts, the session can be terminated or the account can be blocked.

The state-machine approach makes the ATM transaction process easier to understand, implement, test, and maintain. It also helps prevent invalid state transitions and unauthorized operations.

1.2 Need for the Project

Why the Problem Needs to Be Addressed

ATM systems deal directly with financial transactions and sensitive account information. Therefore, security and accurate transaction validation are essential.

An improperly designed ATM system may allow unauthorized users to access accounts, perform invalid transactions, or manipulate transaction processes. A structured state-based approach can reduce these risks by ensuring that each transaction follows a predefined sequence.

The proposed system is needed to:

- Validate ATM cards before allowing transactions.
- Verify the user's PIN securely.
- Prevent unauthorized access.
- Control transaction flow using predefined states.
- Handle incorrect PIN attempts.
- Prevent invalid transaction operations.

- Maintain transaction records.
- Handle transaction failures and session timeouts.
- Improve reliability and security.

Who Is Affected by the Problem

1. **ATM Users** – require secure and reliable banking transactions.
2. **Banking Institutions** – need protection against unauthorized transactions.
3. **System Administrators** – require reliable transaction monitoring.
4. **ATM Service Providers** – need secure and maintainable ATM software.
5. **Bank Employees** – require transaction records for verification and auditing.
6. **Security Teams** – need mechanisms for detecting failed and suspicious attempts.
7. **Software Developers** – require a structured model for implementing ATM operations.

Existing Limitations

Traditional transaction-flow implementations may contain complex conditional statements that become difficult to maintain as the number of transaction conditions increases.

Some systems may also have difficulty clearly representing all possible transaction states and transitions. Improper handling of invalid inputs, failed authentication, session timeout, and transaction errors can create security vulnerabilities.

The proposed state-machine approach provides a structured representation of the ATM workflow and clearly defines the conditions under which state transitions occur.

Engineering Significance

The project demonstrates how **Finite State Machine concepts can be applied to a real-world financial system**. State machines provide a systematic way of representing sequential processes and controlling valid transitions.

The approach improves system reliability by ensuring that operations such as withdrawal or deposit can occur only after successful authentication and account verification.

1.3 Problem Statement

ATM systems must securely authenticate users and correctly manage financial transactions. A transaction should only be allowed when the card, account, and PIN are valid.

The engineering problem addressed in this project is to **design and develop a Smart ATM Transaction Validation System using State Machines to securely control authentication, transaction selection, processing, recording, and completion**.

The system must validate card details, verify the PIN, authenticate the account, allow valid transactions, reject unauthorized operations, and handle failures using appropriate state transitions.

The system should also handle conditions such as incorrect PINs, invalid cards, blocked accounts, insufficient balance, transaction failure, session timeout, and multiple authentication attempts.

1.4 Project Aim

The main aim of this project is to **design and develop a secure ATM transaction validation system using Finite State Machine concepts to control authentication and transaction operations through predefined states and transitions.**

The project aims to improve transaction security, reliability, accuracy, and control while providing a simple and understandable state-based representation of ATM operations.

1.5 Project Objectives

1. To design a state-machine model for ATM transaction validation.
2. To identify and define the major states involved in ATM operations.
3. To implement card and account validation.
4. To implement PIN verification and authentication.
5. To provide secure transaction selection.
6. To support operations such as withdrawal, deposit, and balance enquiry.
7. To validate transactions before processing.
8. To maintain transaction records.
9. To handle incorrect PIN attempts and authentication failures.
10. To handle insufficient balance and transaction failures.
11. To implement session timeout and secure session termination.
12. To test state transitions under valid and invalid conditions.
13. To evaluate system security and reliability.
14. To demonstrate the practical application of state machines in ATM systems.

1.6 Scope

Included in the Project

The project includes:

- Card detection and validation.
- PIN entry and verification.
- Account authentication.
- Transaction selection.
- Cash withdrawal validation.
- Deposit validation.
- Balance enquiry.
- Transaction processing.
- Transaction recording.
- Security and authentication failure handling.
- Multiple PIN attempt handling.
- Session timeout.
- State transition management.
- Testing of valid and invalid transaction conditions.
- Transaction status generation.

Excluded from the Project

The following are outside the primary scope:

- Physical ATM hardware control.

- Real bank account integration.
- Real cash dispenser control.
- Real-time banking network integration.
- Actual financial transactions.
- Biometric authentication hardware.
- ATM camera surveillance.
- Physical card manufacturing.
- Bank infrastructure management.

Assumptions

1. Valid account information is available in the system.
2. The user provides a valid ATM card.
3. PIN information is securely stored.
4. The system receives valid transaction inputs.
5. Account balance information is available.
6. State transitions follow predefined rules.
7. The system is used for educational/project demonstration purposes.

Boundary Conditions

The system operates within the predefined ATM states and transitions. A transaction cannot proceed to processing unless the required authentication and validation states have been successfully completed.

Unauthorized users, invalid cards, incorrect PINs, insufficient balances, and expired sessions are prevented from completing transactions.

1.7 Expected Engineering Outcome

The expected outcome is a **functional state-machine-based ATM transaction validation system**.

1. Authentication System

The system validates the card and PIN before allowing access to transaction services.

2. State Machine Model

The system represents ATM operations using clearly defined states and transitions.

3. Transaction Validation

The system verifies transaction conditions before processing withdrawal, deposit, or balance enquiry operations.

4. Security Mechanism

The system handles incorrect PINs, invalid cards, blocked accounts, and unauthorized access.

5. Transaction Record

Successful and failed transactions are recorded with their corresponding status.

6. Testing

The system is tested using valid and invalid transaction scenarios.

Overall Expected Outcome

The final system demonstrates how state machines can be used to provide **secure, reliable, structured, and controlled ATM transaction processing**.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

A review of existing ATM systems, authentication mechanisms, transaction-processing systems, and finite-state-machine applications was carried out.

The review focused on:

- ATM authentication.
- PIN verification.
- Transaction validation.
- State-machine modelling.
- Banking security.
- Transaction processing.
- Error handling.
- Session management.

Existing approaches were compared based on **security, reliability, complexity, maintainability, validation capability, and implementation suitability**.

2.2 Review of Existing Work

2.2.1 Traditional ATM Systems

Traditional ATM systems provide services such as cash withdrawal, deposit, balance enquiry, and PIN authentication.

These systems generally follow a predefined transaction sequence. However, complex transaction conditions can make the system difficult to model and maintain.

2.2.2 PIN-Based Authentication

PIN authentication is one of the most common methods used by ATM systems. The entered PIN is compared with the stored authentication information.

Incorrect PIN attempts are normally restricted to prevent unauthorized access.

2.2.3 State Machine-Based Systems

Finite State Machines are widely used for systems that involve clearly defined states and transitions.

An ATM is suitable for state-machine modelling because each operation follows a specific sequence.

2.2.4 Transaction Validation

Transaction validation ensures that requested operations satisfy required conditions before they are processed.

For example, a withdrawal request must be checked against account status and available balance.

2.2.5 Security Mechanisms

ATM systems require mechanisms to handle invalid cards, incorrect PINs, blocked accounts, session expiration, and suspicious transaction attempts.

The proposed project incorporates these conditions into the state-transition model.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Traditional ATM System	Provides common banking services	Complex transaction flow	Provides basic ATM functionality
PIN Authentication	Simple and widely used	Vulnerable if credentials are compromised	Used for user verification
Conditional Programming	Easy for small systems	Becomes complex with many conditions	State machine provides better structure
Finite State Machine	Clear states and transitions	Requires proper state design	Highly relevant
Rule-Based Validation	Easy to define transaction rules	Large rule sets can become difficult to manage	Used for transaction validation
Database-Based System	Stores account and transaction data	Requires secure database management	Supports transaction records
Multi-Factor Authentication	Provides stronger security	More complex implementation	Possible future enhancement

2.4 Research / Engineering Gap

Existing ATM systems provide secure transaction services, but representing all possible transaction conditions using conventional programming can become complex.

There is a need for a simple and structured approach that clearly represents:

- Authentication states.
- Transaction states.
- Error states.
- Security states.
- Successful and failed transitions.

The proposed project addresses this gap by applying **Finite State Machine modelling to ATM transaction validation**.

2.5 Proposed Contribution

The project contributes a state-machine-based model for secure ATM transaction validation.

The main contributions are:

1. Development of an ATM state-machine model.
2. Definition of authentication states.

3. Implementation of PIN verification.
4. Transaction validation through state transitions.
5. Handling of failed authentication.
6. Handling of transaction errors.
7. Secure transaction recording.
8. Testing of valid and invalid state transitions.
9. Analysis of security and reliability.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1.1 Stakeholder / User Requirements

Stakeholder	Requirement
ATM User	Secure and easy transaction process
Bank	Secure authentication and transaction processing
System Administrator	Reliable transaction monitoring
Security Team	Prevention of unauthorized transactions
Software Developer	Clearly defined states and transitions
Bank Employee	Access to accurate transaction records
Project User	Simple and understandable ATM interface

3.1.2 Functional and Non-Functional Requirements

Requirement	Type	Measurable Target
Card validation	Functional	Valid cards should be accepted
PIN verification	Functional	Correct PIN should authenticate the user
Account validation	Functional	Valid account should proceed
Transaction selection	Functional	Valid options should be displayed
Withdrawal	Functional	Valid withdrawal should be processed
Deposit	Functional	Valid deposit should be processed
Balance enquiry	Functional	Current balance should be displayed
Transaction recording	Functional	Each transaction should be recorded
Error handling	Functional	Invalid operations should be rejected
Security	Non-Functional	Unauthorized transactions should be prevented
Reliability	Non-Functional	Same input should produce consistent results
Usability	Non-Functional	Interface should be simple to operate
Performance	Non-Functional	Transaction response should occur within a few seconds
Maintainability	Non-Functional	State definitions should be easy to modify

3.2 Constraints & Applicable Standards

Constraints

- System depends on correct account and card information.
- PIN information must be protected.
- The project does not directly connect to a real banking network.
- The system does not control physical ATM hardware.
- Security depends on correct implementation.
- Invalid state transitions must be prevented.
- Transaction records must remain consistent.

Applicable Standards / Guidelines

Standard / Guideline	Requirement	Application
ISO/IEC 25010	Software quality	Reliability, security and usability
ISO/IEC/IEEE 29148	Requirements engineering	System requirements
IEEE 1012	Verification and validation	Testing and verification
ISO/IEC 27001	Information security	Security principles
PCI DSS	Protection of payment-related data	Security reference for sensitive financial information

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Card Validation	Valid card should be accepted	—	High
PIN Attempts	Limited unsuccessful attempts	Count	High
Authentication	Correct PIN required	—	High
Transaction Processing	Only authenticated users	—	High
Withdrawal	Amount must be within available balance	Currency	High
Deposit	Valid amount required	Currency	Medium
Transaction Record	Every transaction should be recorded	—	High
Session Timeout	Session terminates after inactivity	Seconds	Medium
State Transition	Only valid transitions allowed	—	High
Security	Unauthorized transactions prevented	—	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Conditional Programming

Uses if-else statements to control ATM operations.

Advantage: Simple for small applications.

Disadvantage: Becomes difficult to manage when transaction states increase.

Alternative 2: Finite State Machine

Represents every ATM stage as a state with predefined transitions.

Advantage: Clear, structured, and easy to test.

Disadvantage: Requires careful state and transition design.

Alternative 3: Rule-Based System

Uses predefined transaction rules for validation.

Advantage: Flexible validation.

Disadvantage: Large rule sets can become complex.

Evaluation Matrix

Criterion	Weight	Conditional Programming	FSM	Rule-Based
Security	30%	3	5	4
Reliability	20%	3	5	4
Maintainability	20%	3	5	4
Simplicity	15%	5	4	3
Testing	15%	3	5	4

Overall Suitability	100%	3.35	4.80	3.85
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Selected Alternative: Finite State Machine

FSM was selected because it provides a clear representation of ATM states, valid transitions, authentication flow, transaction processing, and error handling.

3.5 Selected Design & Detailed Design

The proposed system uses a **Finite State Machine** to control ATM operations.

Main States

Card Detected



PIN Entry



PIN Verification



Account Verified



Transaction Selection



Transaction Processing



Transaction Recorded



Transaction Completed



Session End

Invalid authentication leads to:

PIN Verification → Authentication Failed → Retry / Session Terminated

System Workflow

Insert Card



Validate Card



Enter PIN



Verify PIN



Verify Account



Select Transaction



Validate Transaction



Process Transaction



Record Transaction



Display Result



End Session

3.6 Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
System model	Conventional flowchart	Simple	Difficult to represent complex states	FSM selected
Authentication	PIN only	Simple	Lower security	PIN validation with attempt control
Transaction control	If-else	Easy initially	Becomes complex	State-based control
Record system	No records	Simple	No transaction history	Transaction records included
Error handling	Generic error	Easy implementation	Less secure	Specific error states
Session control	Manual exit	Simple	Session may remain active	Timeout and termination included

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Efficient software processing	Lightweight state transitions	FSM selected
Ethics	Responsible handling of financial information	Sensitive data protection	Security controls included
Safety	Prevent invalid financial operations	Transaction validation	Invalid transactions rejected
Security	PIN and account protection	Authentication mechanism	Authentication states included
Privacy	Protection of account information	Restricted access	Sensitive information minimized
Reliability	Consistent transaction processing	State validation	Valid transitions enforced

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect PIN	High	Medium	High	Limit attempts	Low
Invalid card	Medium	High	High	Card validation	Low
Unauthorized access	Medium	High	High	Authentication	Medium
Insufficient balance	Medium	Medium	Medium	Balance validation	Low
Invalid transaction	Medium	Medium	Medium	Transaction validation	Low
Session timeout	Medium	Medium	Medium	Automatic termination	Low
Incorrect state transition	Low	High	Medium	State testing	Low
Transaction record failure	Low	High	Medium	Validation and backup	Low
Data modification	Low	High	Medium	Access control	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows a structured development process consisting of requirement analysis, state identification, state-machine design, implementation, testing, and result analysis.

The major stages are:

1. Problem identification.
2. Requirement analysis.
3. ATM state identification.
4. State transition design.
5. System implementation.
6. Transaction validation.
7. Security implementation.
8. Testing and debugging.
9. Result analysis.
10. Documentation.

Resources Used

Resource	Purpose
Computer/Laptop	Development and testing
Programming Language	System implementation
IDE / Code Editor	Coding and debugging
Database	Account and transaction storage
State Diagram Tool	State-machine design
Testing Tools	Functional testing
Documentation Tools	Report preparation

4.2 Hardware / Software / Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Programming Language	System implementation	Development
VS Code / IDE	Coding and debugging	Development
Database	Account and transaction records	Implementation
State Diagram Tool	FSM design	Design
Testing Environment	Test execution	Testing
Computer/Laptop	Running the system	All stages

Implementation Process

1. Create ATM states.
2. Define state transitions.
3. Implement card validation.
4. Implement PIN verification.
5. Implement account validation.
6. Implement transaction selection.
7. Implement transaction processing.
8. Implement transaction recording.
9. Implement error handling.
10. Test all state transitions.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Card validation	Enter valid/invalid card	Correct validation result
PIN verification	Enter correct/incorrect PIN	Correct PIN authenticates
Multiple attempts	Enter wrong PIN repeatedly	Session blocked/terminated
Account validation	Test valid/blocked account	Correct state transition
Withdrawal	Enter valid amount	Transaction processed
Insufficient balance	Enter amount greater than balance	Transaction rejected
Deposit	Enter valid deposit amount	Deposit processed
Balance enquiry	Select balance option	Correct balance displayed
Transaction record	Complete transaction	Record generated
Session timeout	Leave session inactive	Session terminated
Invalid transition	Attempt unauthorized state	Transition rejected

Specification Verification

Specification	Required Value	Achieved Value	Status
Card validation	Valid card accepted	Implemented	Pass
PIN verification	Correct PIN accepted	Implemented	Pass
Failed authentication	Invalid PIN rejected	Implemented	Pass
Transaction validation	Invalid transaction rejected	Implemented	Pass
Transaction recording	Transaction stored	Implemented	Pass
State transitions	Valid transitions only	Implemented	Pass
Session termination	Session ends securely	Implemented	Pass

4.4 Results

The developed Smart ATM Transaction Validation System was tested using different authentication and transaction scenarios.

The system successfully validates cards and PINs before allowing transactions. Correct authentication allows the user to proceed to transaction selection, while incorrect authentication results in an appropriate failure state.

The system also validates transaction conditions such as available balance and transaction amount. Successful transactions are recorded, while failed transactions are rejected and reported appropriately.

Sample Results Table

Test Case	Input	Expected Result	Status
TC01	Valid Card	PIN Entry state	Pass
TC02	Invalid Card	Authentication Failed	Pass
TC03	Correct PIN	Account Verified	Pass
TC04	Incorrect PIN	Retry	Pass
TC05	Multiple Wrong PINs	Session Terminated	Pass
TC06	Valid Withdrawal	Transaction Completed	Pass
TC07	Insufficient Balance	Transaction Failed	Pass
TC08	Balance Enquiry	Balance Displayed	Pass
TC09	Valid Deposit	Deposit Completed	Pass
TC10	Session Timeout	Session Terminated	Pass

4.5 Data Analysis & Engineering Judgment

The testing results demonstrate that the state-machine approach provides a structured method for controlling ATM transactions.

Each operation is performed only when the required previous state has been successfully completed. This reduces the possibility of unauthorized or invalid operations.

The state-machine model also makes it easier to identify errors because every state has predefined valid transitions.

From an engineering perspective, the main advantage of the proposed system is **clarity and controlled transaction flow**. However, the system requires careful state design because missing or incorrect transitions can cause unexpected behavior.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design ATM state machine	State diagram	Fully Achieved
Define ATM states	State definitions	Fully Achieved
Implement card validation	Card validation module	Fully Achieved
Implement PIN verification	Authentication module	Fully Achieved
Implement transactions	Transaction module	Fully Achieved
Record transactions	Transaction record module	Fully Achieved
Handle security failures	Error/security states	Fully Achieved
Test state transitions	Test cases	Fully Achieved
Analyze system performance	Testing results	Fully Achieved
Improve transaction security	Authentication and validation	Fully Achieved

4.7 Strengths & Limitations

Strengths

- Clear state-based transaction flow.
- Improved authentication control.
- Prevents invalid state transitions.
- Easy to understand and test.
- Handles authentication failures.
- Supports transaction validation.
- Maintains transaction records.
- Provides structured error handling.
- Easy to extend with additional states.

Limitations

- Does not connect to a real banking network.
- Does not control physical ATM hardware.
- Does not include biometric authentication.
- Security depends on correct implementation.
- Does not represent all real-world banking conditions.
- Requires careful state and transition design.

4.8 Project Planning, Roles & Budget

Project Milestones

Phase	Activity	Duration
Phase 1	Problem identification	Week 1
Phase 2	Literature review	Week 2
Phase 3	Requirement analysis	Week 3
Phase 4	State-machine design	Week 4
Phase 5	Implementation	Week 5
Phase 6	Testing and debugging	Week 6
Phase 7	Results and analysis	Week 7
Phase 8	Documentation and final review	Week 8

Team Roles

Student	Assigned Responsibility	Actual Contribution
Gunavardhan	Card & User Authentication	Card validation, PIN verification and authentication testing
Pusalapati Chandu	Transaction Processing	Withdrawal, deposit and balance enquiry implementation
Sanjay Soy	Transaction Analysis	Transaction testing, result verification and analysis
Dileep Kumar	Transaction Record & Security	Transaction recording, security controls and error handling

Project Budget

Item	Estimated Cost (₹)	Actual Cost (₹)
Computer/Laptop	0*	0*
Software/IDE	0	0
Database	0	0
State Diagram Tools	0	0
Internet/Data	500	500
Documentation/Printing	500	500
Miscellaneous	300	300
Total	1,300	1,300

*Existing computer resources are assumed.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The **Smart ATM Transaction Validation System Using State Machines** was developed to provide a structured and secure approach to ATM transaction processing.

The project uses a Finite State Machine to represent different ATM operations, including card detection, PIN entry, PIN verification, account validation, transaction selection, transaction processing, transaction recording, and transaction completion.

The system successfully handles valid and invalid authentication attempts and provides appropriate state transitions for different transaction conditions. It also handles insufficient balance, invalid cards, incorrect PINs, transaction failures, and session termination.

The state-machine approach improves the clarity, reliability, security, and maintainability of the ATM transaction process. It ensures that users cannot directly access transaction operations without completing the required authentication states.

Overall, the project demonstrates the practical application of **Finite State Machines in designing secure and reliable financial transaction systems**.

5.2 Future Work

1. Integration with a real banking database.
2. Integration with real ATM hardware.
3. Biometric authentication.
4. OTP-based authentication.
5. Mobile notification for transactions.
6. AI-based suspicious transaction detection.
7. Facial recognition for additional security.
8. Real-time transaction monitoring.
9. Cloud-based transaction management.
10. Advanced fraud detection.
11. Multi-factor authentication.
12. Mobile and web-based ATM monitoring.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

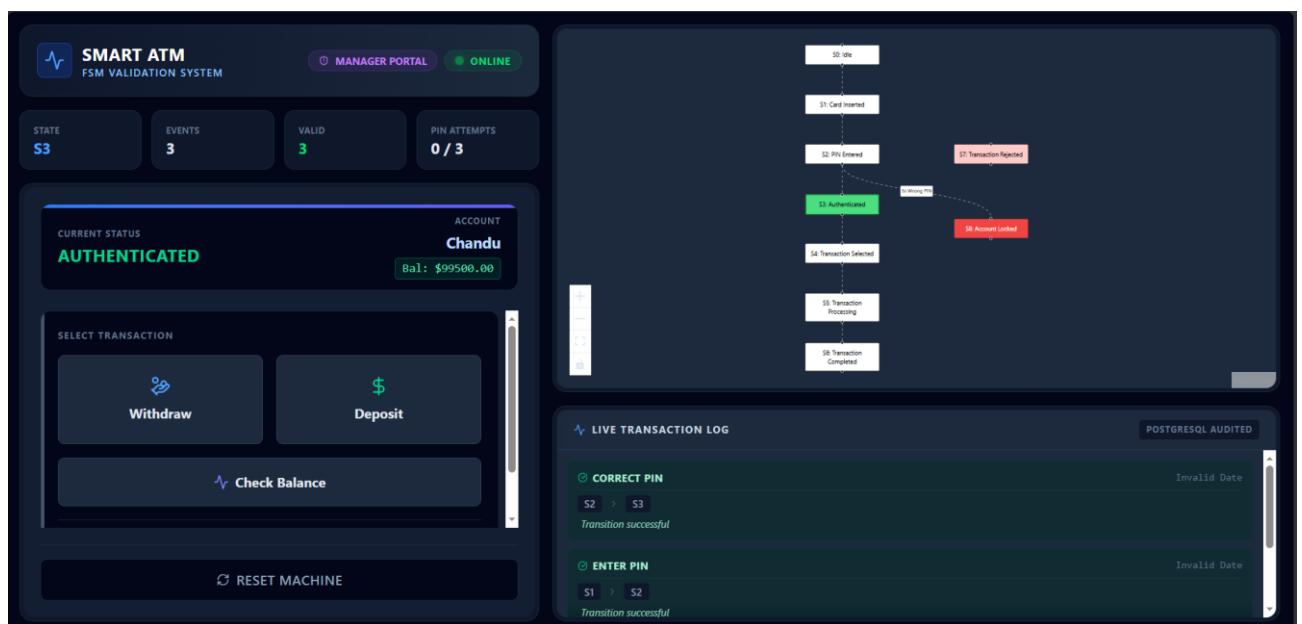
- Finite State Machine concepts.
- ATM transaction processing.
- Authentication mechanisms.
- State transition design.
- Transaction validation.
- Security and error handling.
- Software testing.
- Database-based transaction recording.
- System design.
- Technical documentation.

Engineering & Professional Skills Developed

- Problem identification.
- State-machine modelling.
- Software development.
- Testing and debugging.
- Security analysis.
- Teamwork.
- Project planning.
- Technical communication.
- Engineering decision-making.
- Report preparation.
- Report preparation.

Pusalapati Chandu – Individual Reflection

My main contribution was transaction analysis and testing. I worked on verifying system outputs and analyzing different transaction scenarios. I learned how systematic testing can identify incorrect state transitions and transaction-processing errors.



REFERENCES

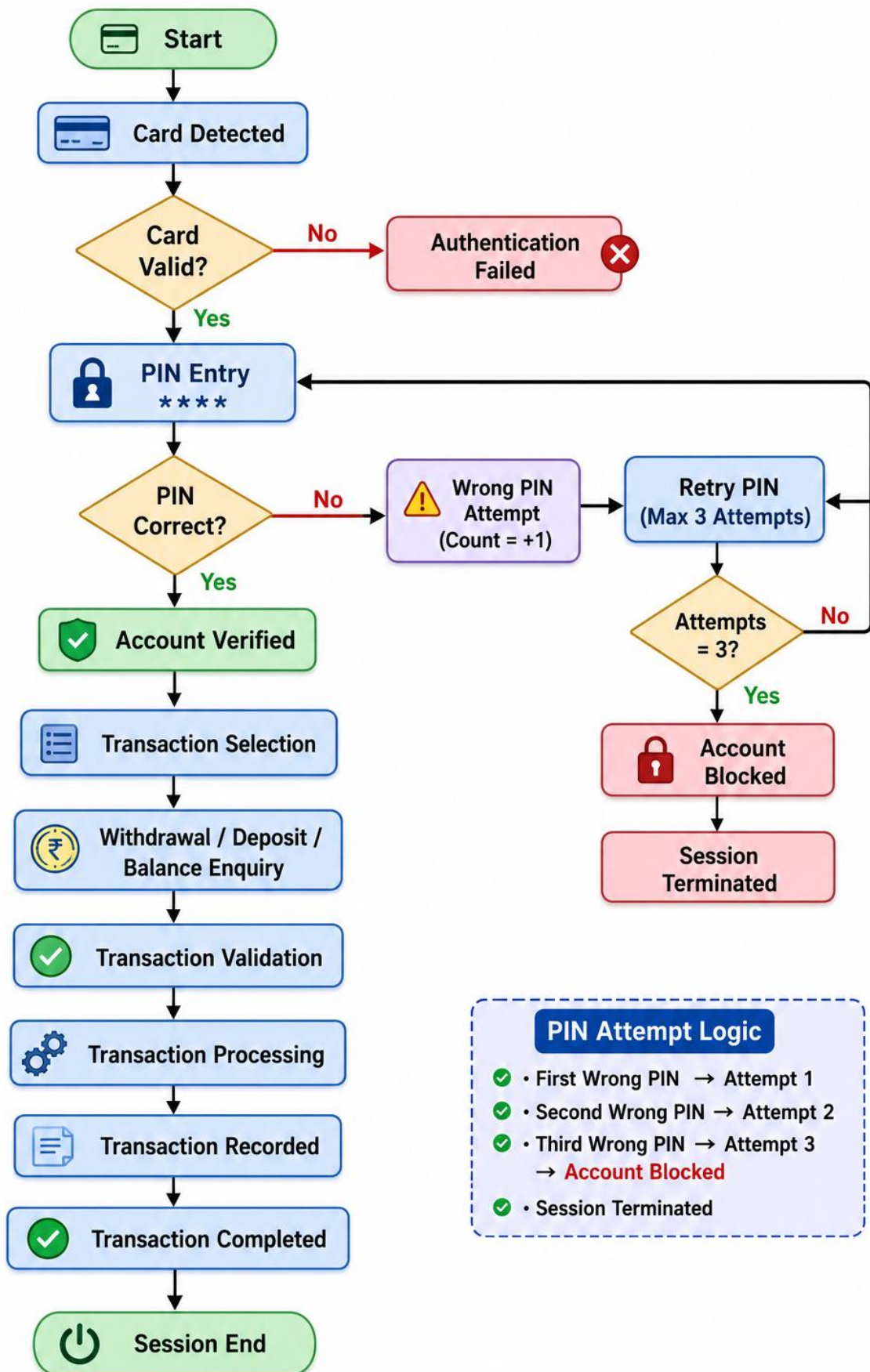
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2. J. E. Hopcroft and J. D. Ullman, *Formal Languages and Their Relation to Automata*.
3. A. V. Aho, M. S. Lam, R. Sethi, and J. D. Ullman, *Compilers: Principles, Techniques, and Tools*.
4. ISO/IEC 25010, *Systems and Software Engineering – Systems and Software Quality Requirements and Evaluation*.
5. ISO/IEC/IEEE 29148, *Requirements Engineering*.
6. IEEE 1012, *System, Software, and Hardware Verification and Validation*.
7. ISO/IEC 27001, *Information Security Management Systems*.
8. PCI Security Standards Council, *Payment Card Industry Data Security Standard*.
9. Documentation of the programming language and database used in the implementation.

APPENDICES

Appendix A – State Transition Table

Current State	Input / Condition	Next State
Card Detected	Valid Card	PIN Entry
Card Detected	Invalid Card	Authentication Failed
PIN Entry	PIN Entered	PIN Verification
PIN Verification	Correct PIN	Account Verified
PIN Verification	Incorrect PIN	Retry / Authentication Failed
Account Verified	Select Transaction	Transaction Selection
Transaction Selection	Withdrawal	Transaction Processing
Transaction Selection	Deposit	Transaction Processing
Transaction Selection	Balance Enquiry	Balance Display
Transaction Processing	Valid Transaction	Transaction Recorded
Transaction Processing	Invalid Transaction	Transaction Failed
Transaction Recorded	Success	Transaction Completed
Transaction Completed	Exit	Session End
Any Active State	Timeout	Session Terminated

Appendix B – ATM State Machine Flowchart



Appendix C – Test Cases

Test Case	Scenario	Expected Result	Status
TC01	Valid card	PIN Entry	Pass
TC02	Invalid card	Reject card	Pass
TC03	Correct PIN	Account verified	Pass
TC04	Wrong PIN	Retry	Pass
TC05	Multiple wrong PINs	Session terminated	Pass
TC06	Valid withdrawal	Withdrawal completed	Pass
TC07	Insufficient balance	Withdrawal rejected	Pass
TC08	Valid deposit	Deposit completed	Pass
TC09	Balance enquiry	Balance displayed	Pass
TC10	Session timeout	Session terminated	Pass

Appendix D – Source Code

The complete source code contains modules for:

- Card validation.
- PIN authentication.
- Account validation.
- State transitions.
- Transaction processing.
- Balance checking.
- Transaction recording.
- Error handling.
- Session management.

Appendix E – Transaction Records

Transaction ID	Account	Transaction Type	Amount	Status
T001	ACC001	Withdrawal	₹2,000	Completed
T002	ACC002	Deposit	₹5,000	Completed
T003	ACC003	Withdrawal	₹10,000	Failed
T004	ACC001	Balance Enquiry	—	Completed

Appendix F – Project Progress Records

Week	Activity	Work Completed	Participants
Week 1	Project Discussion	Problem identified	All members
Week 2	Literature Review	ATM and FSM concepts reviewed	All members
Week 3	Requirement Analysis	System requirements prepared	All members
Week 4	FSM Design	States and transitions designed	All members
Week 5	Implementation	ATM modules developed	All members
Week 6	Testing	Test cases executed	All members
Week 7	Results	Results analyzed	All members
Week 8	Documentation	Final report prepared	All members

Appendix G – Individual Contribution Evidence

Student	Register No.	Major Contribution	Evidence
Gunavardhan	192372041	Card & User Authentication	Authentication module and test cases
Pusalapati Chandu	192372119	Transaction Processing	Transaction module
Sanjay Soy	192311151	Transaction Analysis	Testing and result analysis
Dileep Kumar	192372134	Transaction Record & Security	Security and transaction-record module

Contribution Percentage

Student	Contribution
Gunavardhan	25%
Pusalapati Chandu	25%
Sanjay Soy	25%
Dileep Kumar	25%
TOTAL	100%

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Technical communication	SO3	Whole Report + Viva
Ethics and professional responsibility	SO4	Ch. 3.7
Teamwork and leadership	SO5	Ch. 4.8
Experimentation, testing and engineering judgment	SO6	Ch. 4.3–4.6
Independent acquisition of knowledge	SO7	Ch. 5.3
State-machine/system design	SO2	Ch. 3.3–3.5
Security considerations	Relevant SO	Ch. 3.7
Risk assessment and mitigation	SO2 / SO4	Ch. 3.7
Integrated/system-level engineering	SO1 / SO2	Ch. 3.5
Project planning and management	SO5	Ch. 4.8
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4.8